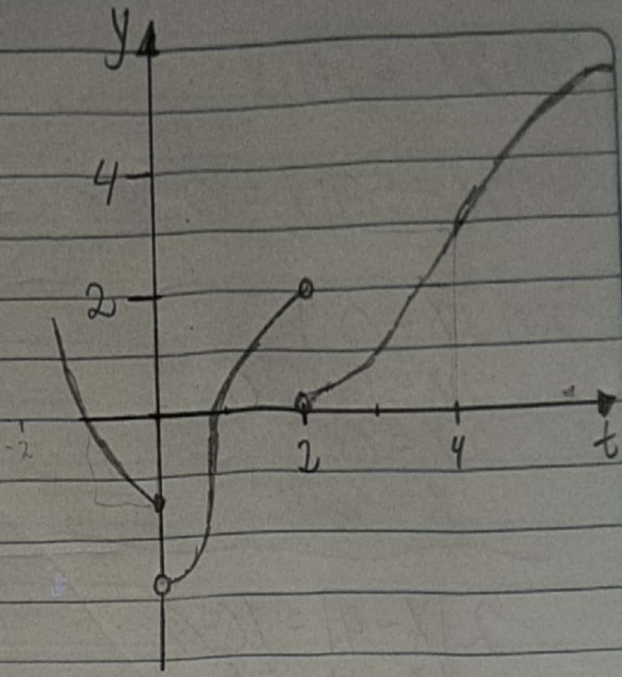
Tarea lección 2.2

7) a) $\lim_{t \to 0^-} g(t) = -1$



b) $\lim_{t \to 0^+} g(t) = -2$

c) $\lim_{t \to 0} g(t) =$ No existe

d) $\lim_{t \to 2^-} g(t) = 2$

e) $\lim_{t \to 2^+} g(t) = 0$

f) $\lim_{t \to 2} g(t) =$ No existe, ya que no se intersectan en el mismo punto

g) $g(2) = 1$

h) $\lim_{t \to 4} g(t) = 3$

9) a) $\lim_{x \to -7} f(x) = -\infty$

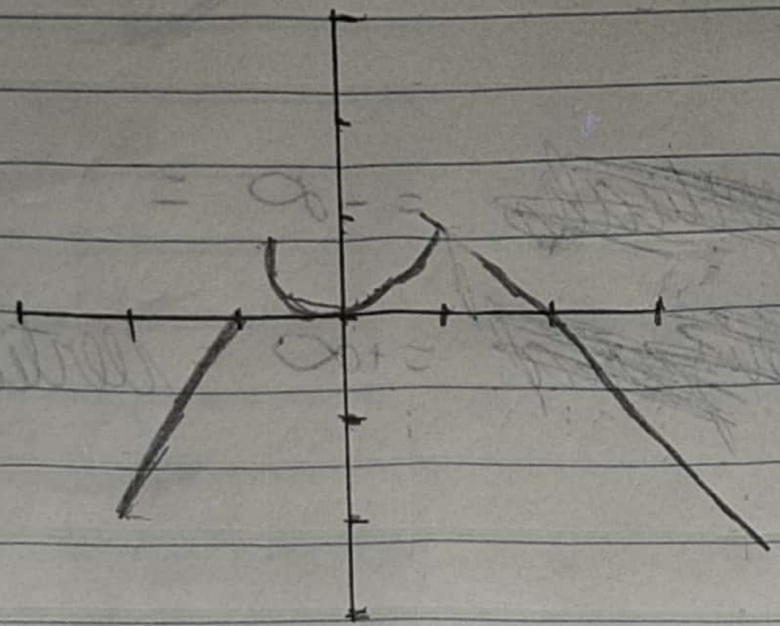
e) $\lim_{x \to 6^+} f(x) = \infty$

b) $\lim_{x \to -3} f(x) = \infty$

c) $\lim_{x \to 0} f(x) = \infty$

d) $\lim_{x \to 6^-} f(x) = -\infty$

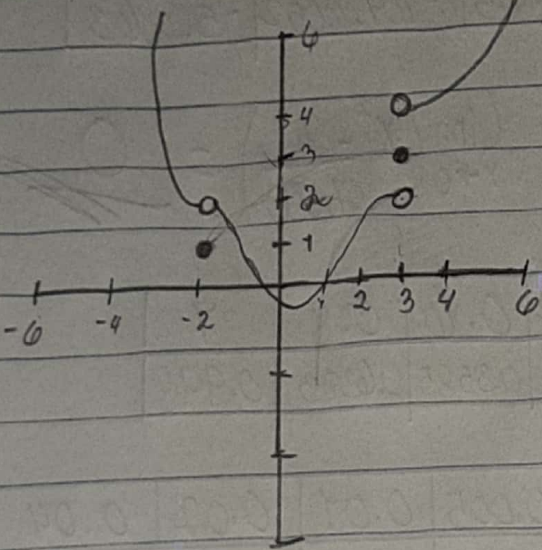
11) $f(x) = \begin{cases} 1 + x & \text{si } x < -1 \\ x^2 & \text{si } -1 \leq x < 1 \\ 2 - x & \text{si } x \geq 1 \end{cases}$



Existe para todas la a menos para -1

17) $\lim_{x \to 3^+} f(x) = 4$, $\lim_{x \to 3^-} f(x) = 2$, $\lim_{x \to -2} f(x) = 2$

$f(3) = 3$, $f(-2) = 1$



27) $\lim_{x \to 0^+} x^x = 1$

0	0.0002	0.0005	0.001	0.0015	0.004	0.01	0.02	0.05	0.1	0.5	0.7	0.9	1
1	0.9982	0.9962	0.9937	0.9903	0.9782	0.9549	0.9247	0.8609	0.7943	0.7071	0.7791	0.9095	1

41) $\lim_{x \to 2^+} = \frac{x^2 - 2x - 8}{x^2 - 5x + 6} = +\infty$

2		2.0001	2.0002	2.0003	2.0004	2.0005	2.001	2.005	2.01	2.1	2.5
	—	80,006	40,006	26,672.67	20,006	16,006.	8,006.01	1,606.03	806.05	86.5	27

43) $\lim_{x \to 0} (\ln x^2 - x^{-2}) = -\infty$

0	0.0002	0.0005	0.05	0.1	0.5
	-2.5×10^{7}	-4.00002×10^{6}	-405.9915	-104.6052	-5.3863

49)

a) $f(x) = x^2 - \left(\frac{2^x}{1000}\right)$ $\qquad \lim_{x \to 0} \left(x^2 - \frac{2^x}{1{,}000}\right) = 0$

0	0.05	0.1	0.2	0.4	0.6	0.8	1
0	0.001465	0.008928	0.0389	0.1587	0.3585	0.6383	0.998

b)

0	0.001	0.003	0.005	0.01	0.02	0.04
	-0.0009997	-0.0009931	-0.0009785	-0.0009070	-0.0006140	0.0005719

$$\lim_{x \to 0} \left(\frac{x^2 - 2^x}{1{,}000}\right) = -0.00099$$